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Fast Watermark Segmentation in LLM generated texts

Collaborators for this talk

This is joint work with



SAYAR KARMAKAR
University of Florida



SUBHRAJYOTY ROY
Wash U STL

Introduction: LLM misuse

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NEWS | 23 September 2025

Journals infiltrated with ‘copycat’ papers that can be written by AI

Tools such as ChatGPT can be used to generate almost-identical research papers that pass standard plagiarism checks. Hundreds are thought to have been published.

1. Introduction

Certainly, here is a possible introduction for your topic: Lithium-metal batteries are promising candidates for high-energy-density rechargeable batteries due to their low electrode potentials and high theoretical capacities [1,2]. However, during the cycle, dendrites forming on the lithium metal anode can cause short circuit, which can affect the safety and life of the battery [3,4]. Therefore, researchers are indeed focusing on various aspects such as negative electrode structure [10], electrolyte additives [11,12], SEI film construction [13,14], and collector modification [15] to inhibit the formation of lithium dendrites. However, using a separator with high mechanical strength and chemical stability is another promising approach to prevent dendrites from infiltrating the cathode. By incorporating a separator with high me-

INVESTIGATIONS ✨ FEATURED

GPTZero finds 100 new hallucinations in NeurIPS 2025 accepted papers

GPTZero's analysis 4841 papers accepted by NeurIPS 2025 show there are at least 100 with confirmed hallucinations



NAZAR SHMATKO, ALEX ADAM, PAUL EBAU
JAN 21, 2026 · 26 MIN READ



RETRACTION

Editor's Note: Retraction of article containing fabricated quotations

We are reinforcing our editorial standards following this incident.

NEW PAPER · FEB 10, 2026 12:09 PM · 1 min

On Friday afternoon, Ars Technica published an article containing fabricated quotations generated by an AI tool and attributed to a source who did not say them. That is a serious failure of our standards. Direct quotations must always reflect what a source actually said.

That this happened at Ars is especially distressing. We have covered the risks of overreliance on AI tools for years, and our written policy reflects those concerns. In this case, fabricated quotations were published in a manner inconsistent with that policy. We have reviewed recent work and have not identified additional issues. At this time, this appears to be an isolated incident.



- **Early approaches:** Exploit differences in cross-entropy/perplexity between human vs AI generated text. [Mitchell et al \(ICML 2023\)](#), [ZeroGPT \(2024\)](#), [Radvand et al \(2025\)](#).

How to detect LLM use?

- **Early approaches:** Exploit differences in cross-entropy/perplexity between human vs AI generated text. [Mitchell et al \(ICML 2023\)](#), [ZeroGPT \(2024\)](#), [Radvand et al \(2025\)](#).
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- **Key Idea: Inject** statistical signals (watermark) into AI-generated text/tokens that can be detected later! ([Christ et al. 2024](#), [Aaronson 2023](#))

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For simplicity, we consider *unwatermarked LLM* $\approx$ Humans.

Watermarking in LLM is everywhere!

Executive Order 14110 of October 30, 2023

Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence

By the authority vested in me as President by the Constitution and the laws of the United States of America, it is hereby ordered as follows:

•
•
•

4.5. *Reducing the Risks Posed by Synthetic Content.* To foster capabilities for identifying and labeling synthetic content produced by AI systems, and to establish the authenticity and provenance of digital content, both synthetic and not synthetic, produced by the Federal Government or on its behalf:

(a) Within 240 days of the date of this order, the Secretary of Commerce, in consultation with the heads of other relevant agencies as the Secretary of Commerce may deem appropriate, shall submit a report to the Director of OMB and the Assistant to the President for National Security Affairs identifying the existing standards, tools, methods, and practices, as well as the potential development of further science-backed standards and techniques, for:

(i) authenticating content and tracking its provenance;

(ii) labeling synthetic content, such as using watermarking;

Watermarking in LLM is everywhere!



« Dismantling Quantum Hype with Tim Nguyen

Google's Sycamore chip: no wormholes, no superfast classical simulation either »

My AI Safety Lecture for UT Effective Altruism

Two weeks ago, I gave a lecture setting out my current thoughts on AI safety, halfway through my year at OpenAI. I was asked to speak by UT Austin's Effective Altruist club. You can [watch the lecture on YouTube here](#) (I recommend 2x speed).

•
•
•

My Projects at OpenAI

So, there are some interesting ideas on the table. The last thing that I wanted to tell you about, before opening it up to Q&A, is a little bit about what actual projects I've been working on in the last five months. I was excited to find a few things that

- (a) could actually be deployed in you know GPT or other current systems,
- (b) actually address some real safety worry, and where
- (c) theoretical computer science can actually say something about them.

I'd been worried that the intersection of (a), (b), and (c) would be the empty set!

My main project so far has been a tool for statistically watermarking the outputs of a text model like GPT. Basically, whenever GPT generates some long text, we want there to be an otherwise unnoticeable secret signal in its choices of words, which you can use to prove later that, yes, this came from GPT. We want it to be much harder to take a GPT output and pass it off as if it came from a human. This could be helpful for preventing academic plagiarism, obviously, but also, for example, mass generation of propaganda—you know, spamming every blog with seemingly on-topic comments supporting Russia's invasion of Ukraine, without even a building full of trolls in Moscow. Or impersonating someone's writing style in order to incriminate them. These are all things one might want to make harder, right?

Watermarking in LLM is everywhere!

How Claude marks content

Claude uses two complementary techniques to mark content generated and processed by Claude: (1) watermarks embedded in text, and (2) signed provenance metadata attached to files.

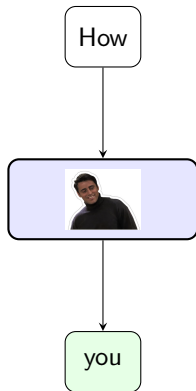
1. Embedded watermarks in text

When a supported Claude model generates text, it weaves an imperceptible watermark directly into the text itself. You won't see it, and it doesn't change the meaning, quality, or readability of Claude's response.

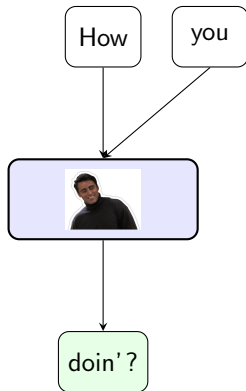
Because the watermark is part of the text, it will travel with the text when it's copied and pasted elsewhere, and may persist through some editing. Watermarking will be applied at the model level, which means it will be present no matter which Claude product or surface the text comes from.

How does watermarking work (roughly)?

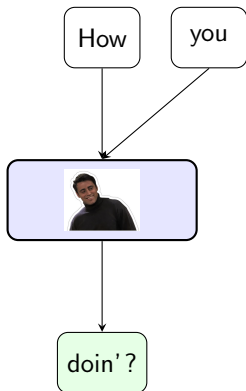
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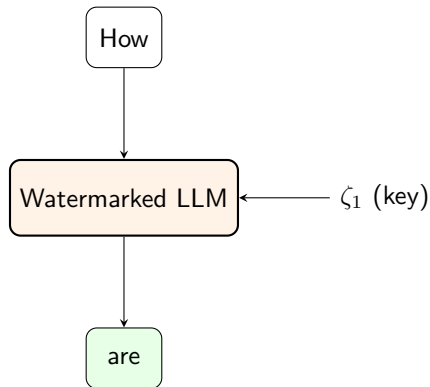
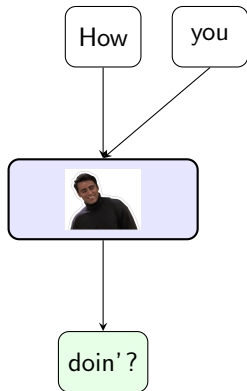


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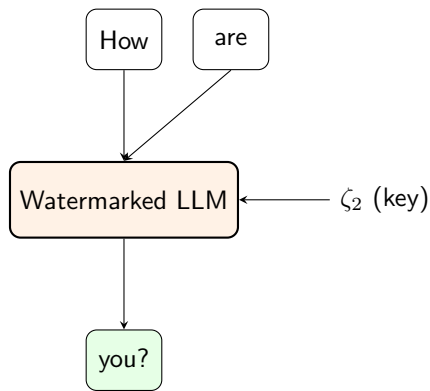
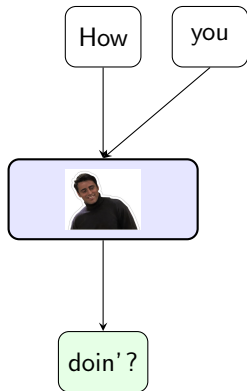


Given tokens $w_1, w_2, \dots, w_{t-1}$, the unwatermarked LLM or human implicitly computes a next token probability (NTP) distribution P_t over the vocabulary $\mathcal{W}$. The next token is chosen as $w_t \sim P_t$.

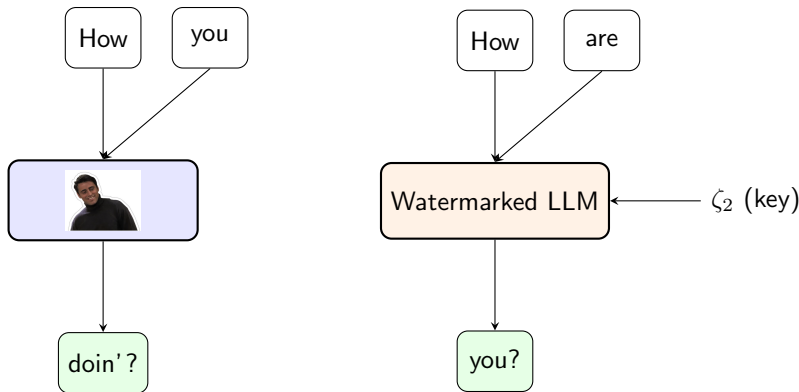
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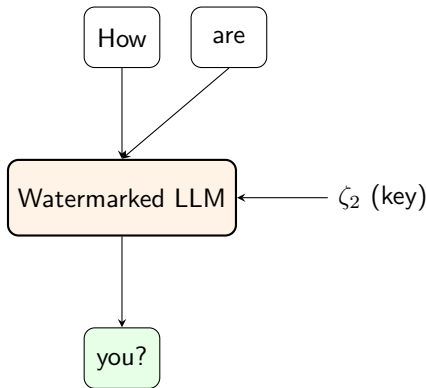
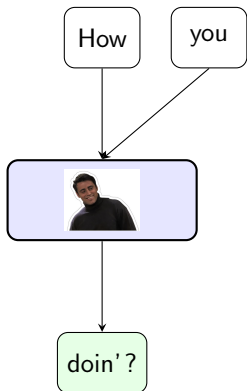


How does watermarking work (roughly)?



Given tokens $w_1, w_2, \dots, w_{t-1}$, the watermarked LLM computes NTP P_t , and the next *watermarked* token is chosen as $w_t \sim S(P_t, \zeta_t)$. Here $S \leftarrow$ decoder function, $\zeta \leftarrow$ keys.

How does watermarking work (roughly)?



The keys ζ_t are *pseudo-random*: they can be constructed from a given text via cryptographic algorithms, and ζ_t are **i.i.d.**. We will assume ζ_t are known to the verifier.

Example: Gumbel-max Watermarking¹

- 1 Assume finite dictionary $\mathcal{W}$. Use LLM to produce NTP P_t on $\mathcal{W}$.
- 2 Generate i.i.d. random variables $U_1, U_2, \dots, U_{|\mathcal{W}|} \sim U(0, 1)$.
- 3 Output

$$w_t = \arg \max_{w \in \mathcal{W}} \frac{\log(U_w)}{P_t(w)}$$

- 4 Gumbel-max trick: w_t still follows² P_t . An example of *unbiased* watermarking.

¹Aaronson, 2023. <https://www.youtube.com/watch?v=2Kx9jbSMZqA>

²Homework for anybody interested

Example: Gumbel-max watermarking

Legend: Unwatermarked Gumbel Watermark

It is possible more people who came in contact with a man who died from the Ebola virus in Dallas, Texas, could test positive for the virus in the coming days, the head of the Centers for Disease Control and Prevention said Sunday. Tomohiro Mori, director of the CFDA'S Washington state office, confirmed to CNN that three cases of Ebola North American have been confirmed in the Dallas-area. Two of those echoe since 24 Saturday were transferred to Democratic Texas state hospitals because of random sneezing, said Margaret Summers of the Farrand Health System in Washington. One patient who traveled to New York â– a man from Japan whose travel had presumably been made by the Ebola vaccine but who traveled all from Japan

Figure: Example paragraph where tokens 70 – 100 has been watermarked.

Key problem addressed:

$H_0 : w_{1:n}$ is unwatermarked

$H_1 : w_{1:n}$ has watermarked patches (from a particular fixed watermarking

How do they test?

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- NTP's P_t are unknown. However, under H_0 of no watermarking, there exists a score function h such that $Y_t := h(w_t, \zeta_t)$ are i.i.d. with known mean μ_0 .

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- Example for *Gumbel-max* Pivot Statistics:
 $Y_t = -\log_2(1 - U_{t,\omega_t})$. If ω_t is not watermarked, $Y_t \sim \text{Exp}(1)$.
If ω_t is watermarked, Y_t is very large. Note that $\mu_0 = 1$.

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- *Elevated alternatives yield power:*

For a good¹ h , usually $\inf_{P \in \mathcal{P}} \mathbb{E}_{1,P}(Y_t) \geq \mu_0 + d$, where $d > 0$ is constant and $\mathbb{E}_{1,P}$ denotes expectation under watermarking when the NTP is P . $\mathcal{P}$ is some class of possible NTPs. This means that under watermarking, distribution of Y_t is well-separated from its null distribution.

¹Good choice of h depends on the watermarking scheme which is usually known to the verifier. The choice of optimal h : [Cai et al. 2024](#), [Kuditipudi et al. 2024](#), [Li, Ruan, Wang, Long & Su 2025 a](#) and [b](#), [Cai et al. 2025](#).

Example of elevated alternative

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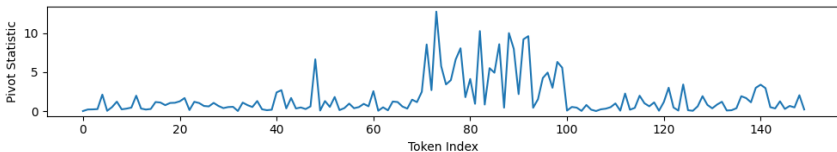


Figure: [Top]: Example Sentence where tokens 70 – 100 has been watermarked. [Bottom]: Corresponding pivot statistic.

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Existing approaches Li et al. (2024), Zhao, Liao, Wang & Li (2024), Pan et al. (2025),:

- Mostly slow algorithms.
- No general theoretical guarantees.

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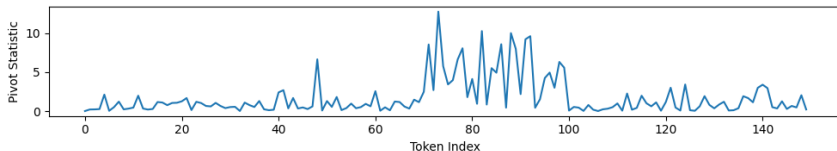
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This gap is also pointed out by [Li, Wen, He, Wu, Long & Su \(2025\)](#).

Goal: Give a computationally feasible + theoretically consistent algorithm to identify watermarked patches in a text.

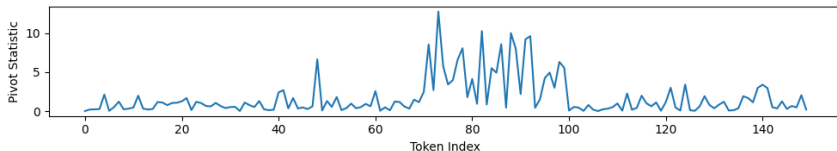
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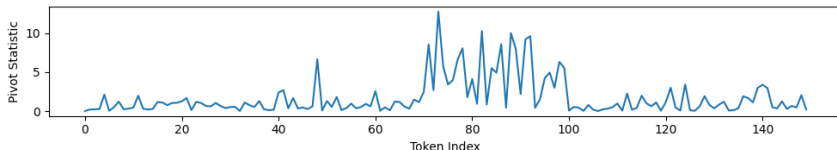
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National Institutes of Health (.gov)

<https://pubmed.ncbi.nlm.nih.gov> > ...

The Cusum Test of Homogeneity With an Application in ...

by B Levin · 1985 Cited by 147 – We test the hypothesis of homogeneity over time in the rate of an event with a modification of Page's cumulative sum (cusum) procedure.

Key Challenges

Not an usual mean-based change-point problem, since $\mathbb{E}_{1,P_t}[Y_t]$ changes with t , and depends on P_t and is unknown.

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Two key challenges.

- What will be our algorithmic approach to fuse computational speed and statistical power?
- Existing sporadic epidemic change-point papers focus on stationary or i.i.d. data (Hušková (1995), Csörgö & Horváth (1997), Chen et al. (2016)). Y_t are i.i.d. under no watermarking, but not when ω_t is watermarked.

If there exists *only one* watermarked interval I , then estimate it as:

$$\hat{I} = \arg \min_{s,t \in [n]} \sum_{k \notin [s,t]} (Y_k - \mu_0 - \rho \tilde{d}),$$

where $\rho \in (0, 1)$ and $\tilde{d}$ must satisfy² $\tilde{d} > d > 2\rho\tilde{d}$.

²Recall d is such that $\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{1,\mathbb{P}}(Y_t) \geq \mu_0 + d$.

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Theorem (Simplified)

Under above assumptions, $|\hat{I} \Delta I| = O_{\mathbb{P}}((\rho\tilde{d})^{-1})$

- *Maximum subarray problem*: can be computed in $O(n)$ using Kadane's algorithm.

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- What if there are multiple patches?

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If number of watermarked patches, K , is **known**, use:

$$(\hat{I}_1, \dots, \hat{I}_K) = \arg \min_{J_1 < J_2 < \dots < J_K} \sum_{k \notin \cup_{i=1}^K J_i} (X_k - \mu_0 - \rho \tilde{d}),$$

What about its theoretical properties?

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Assumption 0: (*Min. separation and size*) Each true watermarked interval $\{I_j\}_{j=1}^K$ has size at least $\log n$, and they are separated by at least $\log n$ distance.

Theorem (Simplified)

Under above assumptions, $\sum_{j \in [K]} |\hat{I}_j \Delta I_j| = O_{\mathbb{P}}((\rho d)^{-1})$.

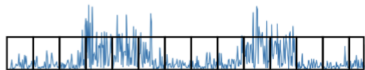
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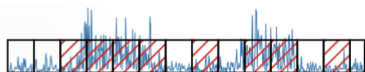
How to identify multiple watermarked patches when the number of patches is unknown?

WISER: Watermark Identification via Segmenting Epidemic Region

- 1 Given $\{Y_t\}_{t=1}^n$, create blocks of length b . Calculate block sums $S_1, \dots, S_{\lceil n/b \rceil}$.



Take blocks of size b and calculate the block sums.



The block sums of red shaded blocks exceed Q

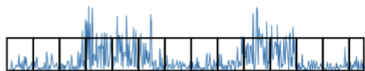
To determine Q , take $(1-\alpha)$ -th quantile of null distribution of maximum block sums

^aUnder no watermarking, Y_t are i.i.d and their distribution is known.

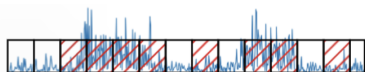
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- 2 Keep blocks where $S_i > Q$. Here Q is $(1 - \alpha)$ -th quantile^a of null (i.e under no watermarking) distribution of maximum block sum $\max_i |S_i|$.

^aUnder no watermarking, Y_t are i.i.d and their distribution is known.



Take blocks of size b and calculate the block sums.

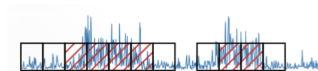


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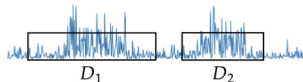
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WISER: Watermark Identification via Segmenting Epidemic Region

- 3 Join the selected consecutive blocks. The smaller, “lone wolf” boxes must be spurious. Hence, discard blocks smaller than $cn^{1/p}$ for some $p > 2$.



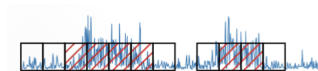
Form connected intervals based on selected blocks.
Remove the connected intervals with relatively small lengths.



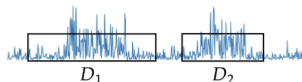
Slightly enlarge the remaining connected intervals
to form intervals D_i .

WISER: Watermark Identification via Segmenting Epidemic Region

- 3 Join the selected consecutive blocks. The smaller, “lone wolf” boxes must be spurious. Hence, discard blocks smaller than $cn^{1/p}$ for some $p > 2$.
- 4 Slightly enlarge each block (by $c\sqrt{\log n}$ on both sides), call them D_i .



Form connected intervals based on selected blocks.
Remove the connected intervals with relatively small lengths.



Slightly enlarge the remaining connected intervals
to form intervals D_i .

Recall: If there exists *only one* watermarked interval I , then estimate it as:

$$\hat{I} = \arg \min_{s,t \in [n]} \sum_{k \notin [s,t]} (Y_k - \mu_0 - \rho \tilde{d}), \quad \rho \in (0,1), \quad \tilde{d} > d > 2\rho\tilde{d}.$$

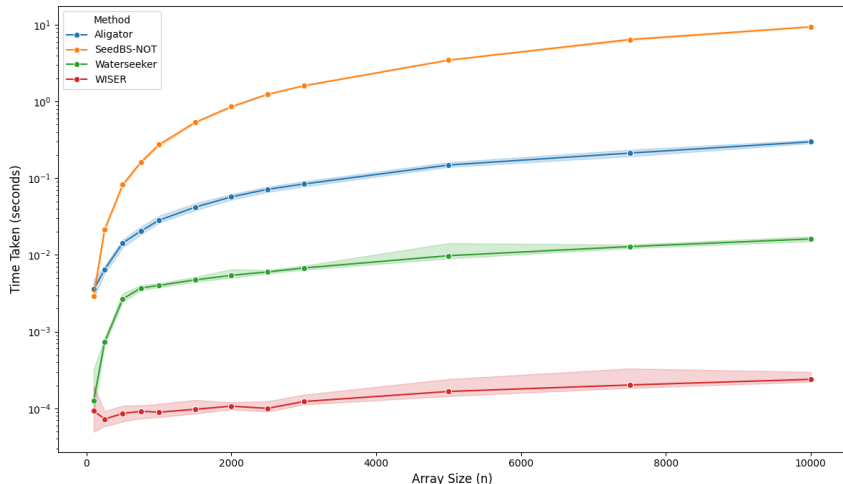
- 5 Each D_i covers exactly *one* watermarked patch, and the D_i 's are disjoint. Use the single watermark patch estimate on each detected block interval D_i . Final estimates $\leftarrow \{\hat{I}_j\}_{j=1}^{\hat{K}}$.

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- 6 Estimate $\tilde{d}$ using average of $Y_t - \mu_0$ over detected blocks, i.e., over $\cup_{i=1}^{\hat{K}} D_i$.
- 7 Detailed ablation studies provide practically-informed choices of ρ and block size b .

Time Complexity



Time complexity of WISER is $O(n)$.

Aligator: [Li et al. \(2024\)](#), SeedBS-Not: [Zhao, Liao, Wang & Li \(2024\)](#),
Waterseeker: [Pan et al. \(2025\)](#).

$K \leftarrow$ the number of true watermarked segments.

Assumption 1: (*Min. separation and size*) Each true watermarked interval $\{I_j\}_{j=1}^K$ have size at least n^ν , and they are separated by at least $n^\nu \log n$ distance (*compared to $\log n$ for the oracle estimator*).

If $\mathbb{E}_0[|X - \mu_0|^p] < \infty$ then $\nu > 1/p$ suffices.

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Assumption 2: $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{1,\mathbb{P}}(Y) < \infty$, $d \geq c$ (watermarking schemes do not change with increasing text-length).

Theorem (Simplified)

Under above assumptions, as $n \rightarrow \infty$,

$$\mathbb{P} \left(\hat{K} = K; \min_{1 \leq j \leq K} |I_j \Delta \hat{I}_j| < \frac{M}{d} \right) \rightarrow 1.$$

- The higher the degree of separation d , the more the localization accuracy.

Benchmark Results (Gumbel)

Model Name	Method	IOU	Precision	Recall	F1	RI	MRI
facebook opt-1.3b (50272)	Oracle	0.944 (0.71-0.99)	0.998	0.998	0.998	0.985 (0.92-1.00)	0.984 (0.92-1.00)
	WISER	0.921 (0.72-0.99)	1.000	0.995	0.997	0.976 (0.92-1.00)	0.974 (0.90-1.00)
	Aligator	0.514 (0.03-0.88)	0.245	0.930	0.388	0.894 (0.67-0.97)	0.874 (0.63-0.97)
	SeedBS	0.327 (0.00-0.86)	0.623	0.445	0.519	0.761 (0.20-0.96)	0.710 (0.14-0.96)
	WaterSeeker	0.657 (0.37-0.92)	1.000	0.807	0.893	0.860 (0.70-0.98)	0.846 (0.66-0.98)
facebook opt-125m (50272)	Oracle	0.951 (0.83-0.99)	1.000	1.000	1.000	0.987 (0.96-1.00)	0.987 (0.95-1.00)
	WISER	0.934 (0.81-0.99)	1.000	0.995	0.997	0.980 (0.94-1.00)	0.979 (0.94-1.00)
	Aligator	0.745 (0.29-0.95)	0.410	0.985	0.579	0.940 (0.85-0.99)	0.933 (0.81-0.99)
	SeedBS	0.500 (0.00-0.95)	0.760	0.630	0.689	0.844 (0.52-0.99)	0.811 (0.46-0.99)
	WaterSeeker	0.672 (0.38-0.93)	1.000	0.802	0.890	0.864 (0.70-0.98)	0.850 (0.66-0.98)
google gemma-3-270m (262144)	Oracle	0.916 (0.14-0.99)	0.980	0.980	0.980	0.976 (0.75-1.00)	0.974 (0.70-1.00)
	WISER	0.903 (0.00-0.99)	0.965	0.960	0.962	0.953 (0.20-1.00)	0.950 (0.14-1.00)
	Aligator	0.539 (0.00-0.87)	0.252	0.920	0.395	0.881 (0.20-0.97)	0.863 (0.14-0.97)
	SeedBS	0.299 (0.00-0.86)	0.562	0.420	0.481	0.727 (0.20-0.96)	0.677 (0.14-0.96)
	WaterSeeker	0.646 (0.12-0.93)	0.973	0.780	0.866	0.854 (0.69-0.98)	0.838 (0.65-0.98)
meta-llama Meta-Llama-3-8B (128256)	Oracle	0.940 (0.74-0.99)	0.995	0.995	0.995	0.982 (0.94-1.00)	0.981 (0.93-1.00)
	WISER	0.928 (0.69-0.99)	1.000	0.988	0.994	0.977 (0.91-1.00)	0.975 (0.90-1.00)
	Aligator	0.572 (0.01-0.92)	0.379	0.940	0.540	0.915 (0.71-0.98)	0.897 (0.66-0.97)
	SeedBS	0.383 (0.00-0.98)	0.645	0.510	0.570	0.768 (0.20-0.99)	0.720 (0.14-0.99)
	WaterSeeker	0.570 (0.33-0.91)	1.000	0.720	0.837	0.814 (0.67-0.98)	0.791 (0.63-0.98)
mistralai Mistral-7B-v0.1 (32000)	Oracle	0.920 (0.68-0.99)	0.998	0.998	0.998	0.978 (0.92-1.00)	0.977 (0.92-1.00)
	WISER	0.880 (0.59-0.98)	1.000	0.998	0.999	0.965 (0.87-1.00)	0.961 (0.84-1.00)
	Aligator	0.310 (0.00-0.86)	0.226	0.750	0.348	0.823 (0.55-0.96)	0.788 (0.48-0.96)
	SeedBS	0.220 (0.00-0.79)	0.405	0.278	0.329	0.611 (0.20-0.95)	0.561 (0.14-0.94)
	WaterSeeker	0.621 (0.36-0.88)	1.000	0.765	0.867	0.840 (0.69-0.97)	0.824 (0.65-0.97)
princeton-nlp Sheared-LLaMA-1.3B (32000)	Oracle	0.945 (0.81-0.99)	1.000	1.000	1.000	0.985 (0.95-1.00)	0.985 (0.94-1.00)
	WISER	0.928 (0.75-0.99)	1.000	0.998	0.999	0.979 (0.94-1.00)	0.978 (0.93-1.00)
	Aligator	0.485 (0.01-0.90)	0.237	0.930	0.378	0.890 (0.61-0.97)	0.868 (0.55-0.97)
	SeedBS	0.302 (0.00-0.91)	0.512	0.388	0.441	0.728 (0.20-0.98)	0.678 (0.14-0.98)
	WaterSeeker	0.659 (0.38-0.94)	1.000	0.812	0.897	0.862 (0.70-0.98)	0.847 (0.66-0.98)

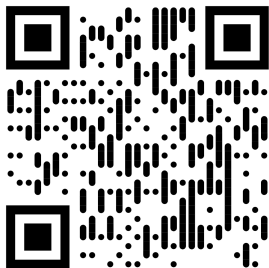
Figure: News articles (500 words) with watermarked patches at 100–200 and 325–400. (...) indicate 95% coverage intervals based on 500 Monte-Carlo draws.

- What if watermarks do not occur in intervals, but say at every alternate token? Use DCART-type approach [Padilla, Yu, Rinaldo \(2021\)](#).
- What if one text uses many watermarking schemes ($\approx$ many LLMs)? Both testing and estimation is unknown, but recent works on watermarking and e-values [Su, Wang, Zhao, \(2026\)](#); [Huang et. al. \(2026\)](#) may provide answers.

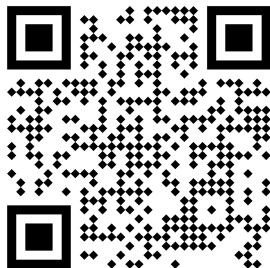
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Thank you



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Questions?

Examples of elevated alternatives

Suppose $\mathcal{P}_\Delta := \{\max_{w \in \mathcal{W}} P_w \leq 1 - \Delta\}$ for some fixed $\Delta > 0$. Then, for Gumbel-watermarking, it follows that

$$\begin{aligned} & \inf_{\mathbb{P} \in \mathcal{P}_\Delta} \mathbb{E}_{1, \mathbb{P}}[h(Y)] \\ & \geq \sum_{n=1}^{\infty} \left(\frac{1}{n} - \left\lfloor \frac{1}{1-\Delta} \right\rfloor \frac{(1-\Delta)^2}{1+n(1-\Delta)} - \frac{(1 - (1-\Delta)\left\lfloor \frac{1}{1-\Delta} \right\rfloor)^2}{1+n(1 - (1-\Delta)\left\lfloor \frac{1}{1-\Delta} \right\rfloor)} \right). \\ & := d + \mu_0 = d + 1. \end{aligned}$$

Δ	d
1/2	0.500000
1/4	0.417220
1/5	0.376476